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Laboratory #4: CMOS Inverter Layout

I. OBJECTIVES

- To develop skills for CMOS layout using Synopsys Custom Compiler. • To create an inverter layout and verify it using DRC and LVS.
- To perform parasitic extraction and post-layout simulations.

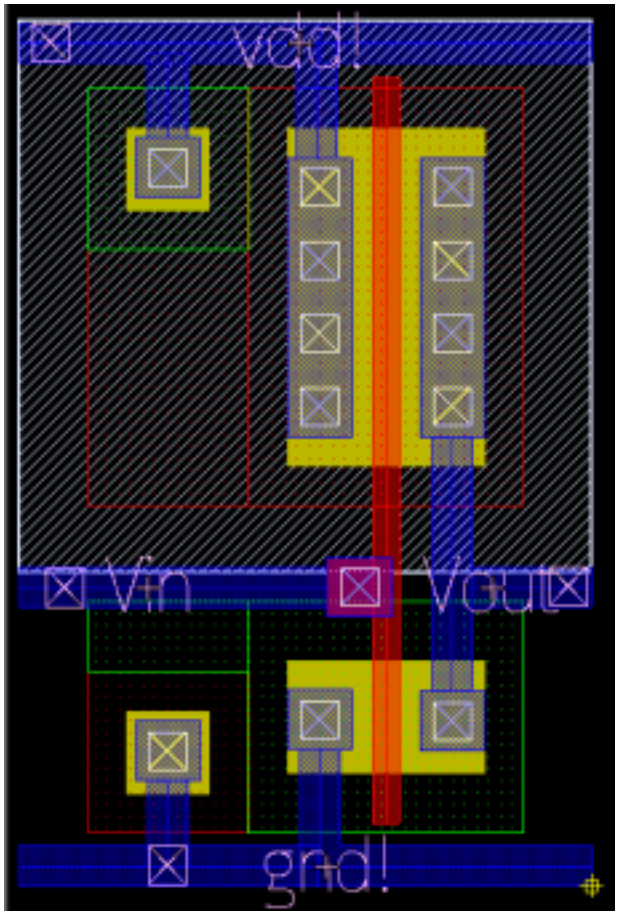
II. LAB PROCEDURE

1. Inverter Layout

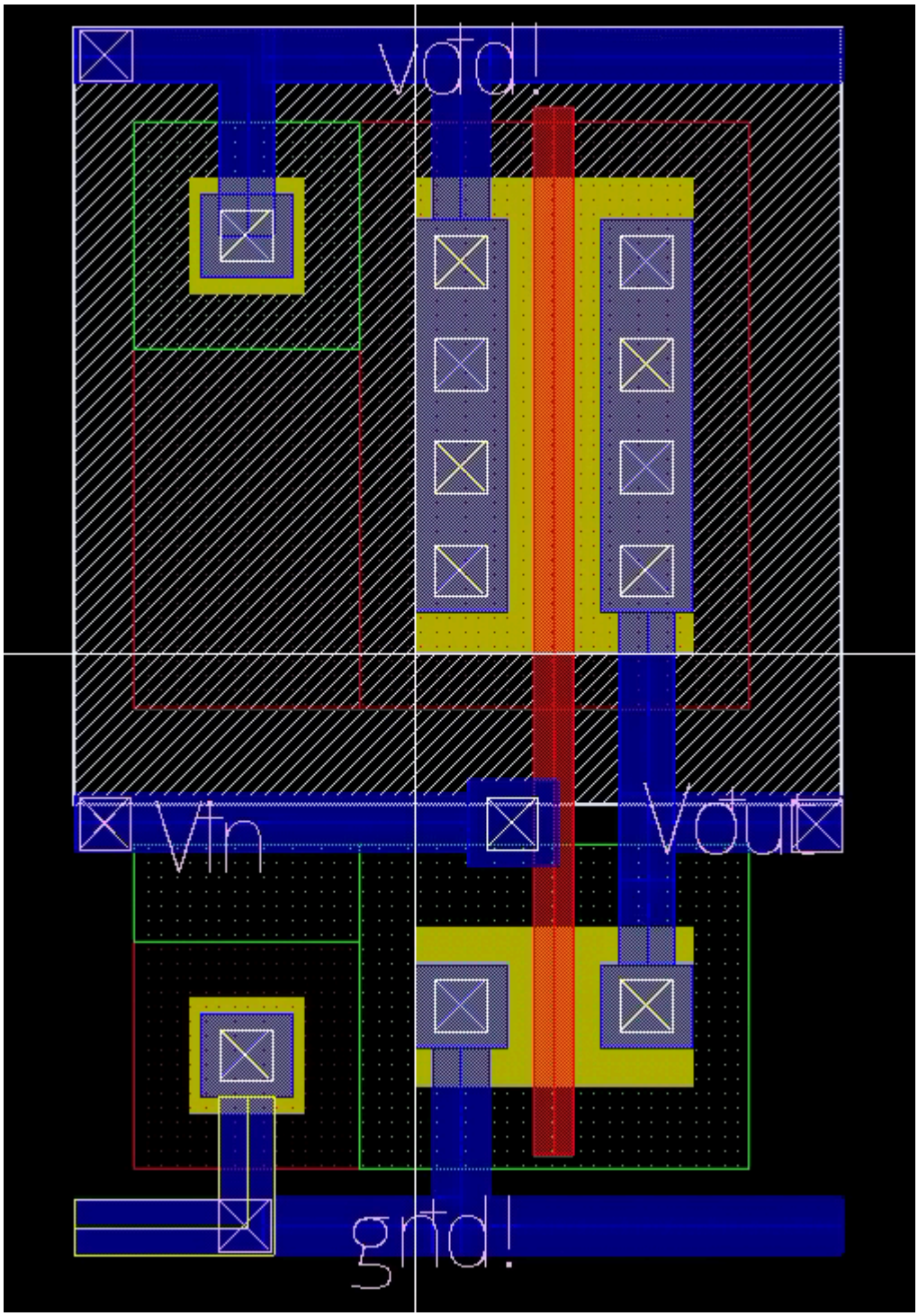
In lab 2, an inverter schematic and symbol were created. In this lab assignment, we will create a layout for that same inverter.

- Re-open the tutorial library where you created the inverter schematic/symbol in the previous assignment.
- Follow the Synopsys Tutorial Part 2, pages 1-20 to complete the layout of the inverter.

Sample



My Actual Layout



Question:

1) For your lab report, include a table of all the layout layers along with contact layers you used, and their corresponding names in Synopsys tools (ex. Metal 1 : M1)

Layout Layers & Contact Layers	**Corresponding Name**
Diffusion	DIFF
P-type implant	PIMP
N-type implant	NIMP
Metal 1	M1
Polysilicon	PO
Substrate	NWELL

2) Mention the layer that provide PMOS & NMOS channel length and width dimensions. Looking at 2D image of the layout, mention the axis that determines Length and Width of the transistors. (ex. X-axis : _____)

X-axis determines channel length from polysilicon layer.

Y-axis determines channel width from diffusion layer.

2. DRC and LVS

To verify that the layout is correct, a Design Rules Check (DRC) must be performed to verify that the layout complies with design rules for this 90nm process. Furthermore, a Layout vs. Schematic (LVS) check must be performed to verify that layout you created actually matches the schematic of the desired inverter.

- Follow the Layout Verification section in your tutorial, pages 21-28 to perform DRC and LVS. Adjust your layout if necessary to fix any errors. The tutorial includes a troubleshooting guide.

VUE: [...sers/student/dthornburg/ELEN153L/synopsys_custom/pvjob_Lab3.Inverter_Creation.icv.drc/Inverter_Creation.vue]

File View Tools Windows Help

Load Results × | Run Summary ×

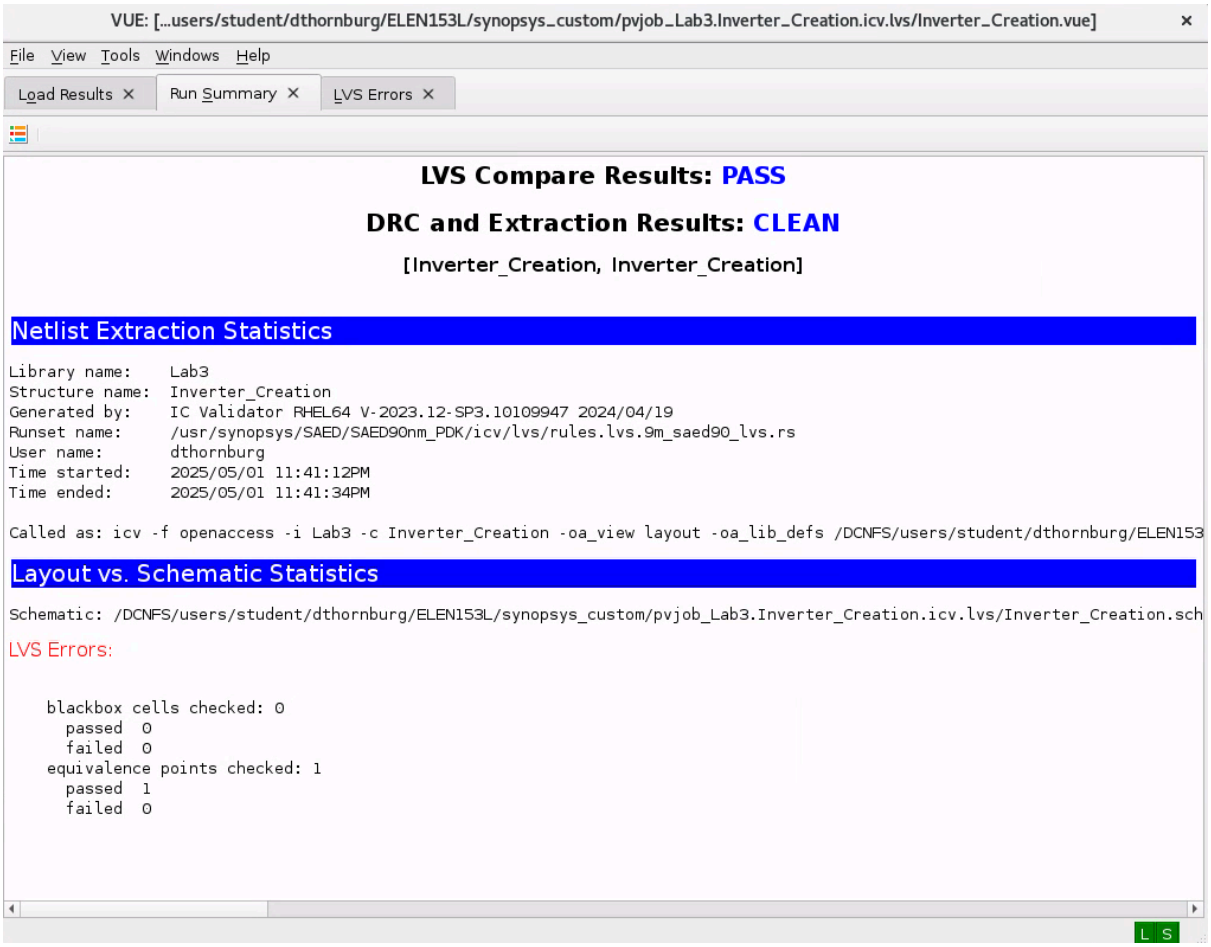
LAYOUT ERRORS RESULTS

CLEAN

DRC Error Statistics

```
Library name:    Lab3
Structure name:  Inverter_Creation
Generated by:    IC Validator RHEL64 V-2023.12-SP3.10109947 2024/04/19
Runset name:     /usr/synopsys/SAED/SAED90nm_PDK/icv/drc/rules.drc.9m_saed90_icv.rs
User name:       dthornburg
Time started:    2025/05/01 11:40:33PM
Time ended:      2025/05/01 11:40:38PM

Called as: icv -f openaccess -i Lab3 -c Inverter_Creation -oa_view layout -oa_lib_defs /DCNFS/users/student/dthornburg/ELEN153
```



III. REPORT

Write a short laboratory report that details all the work done. Describe the objective and procedures of this lab with your own words. The lab report should contain the following:

a) All schematics and layouts used in your lab.
b) Screenshots showing your design passed DRC and LVS
c) Answer any questions in the lab assignment
d) Conclusions